

Algorithm Brainstorming					Algorithm Selection				
Criteria	Weight	Insertion Sort	Cycle Sort	Gnome Sort	Criteria	Weight	Insertion Sort	Cycle Sort	Gnome Sort
Conceptually different	5	0	+	0	Stable	4	0	-	0
Easy to code from scratch	5	0	-	0	In-place	5	0	0	0
Easy to understand and explain	4	0	0	0	Minimises number of swaps	5	0	+	-
Suitability for non-numeric data	4	0	0	0	Suitability for large datasets	4	0	0	0
Length of code	2	0	-	0	Average and worst case Time Complexity	2	0	0	0
Sum of “+” (Weight)(+)		0	1x5 = 5	0	Sum of “+” (Weight)(+)		0	1x5 = 5	0
Sum of “0” (Weight)(0)		5x0 = 0	1x0 = 0	5x0 = 0	Sum of “0” (Weight)(0)		5x0 = 0	3x0 = 0	4x0 = 0
Sum of “-” (Weight)(-)		0	1x(-5) + 1x(-2) = -7	0	Sum of “-” (Weight)(-)		0	1x(-4) = -4	1x(-5) = -5
Net (out of 20)		0	5 + 0 + (-7) = -2	0	Net (out of 20)		0	5 + 0 + (-4) = 1	0 + 0 + (-5) = -5
https://www.geeksforgeeks.org/dsa/insertion-sort-algorithm/					https://www.geeksforgeeks.org/dsa/cycle-sort/				
					https://www.geeksforgeeks.org/dsa/gnome-sort-a-stupid-one/				

User Interface

Sign in

Register

Exercise 1

Exercise 2

Exercise 3

Product Spreadsheet

Welcome! Add products by keying in the information and then click add

Enter ID:

Enter Name:

Enter Price:

Add

Sort by ID

Sort by Name

Sort by Price

Clear

	ID	Name
0		
1		

Product Gallery

Brainstorming for data

These systems manage people, tasks and operations.

Examples:

- Hospitals
- Airports
- Banks
- Customer service centers
- Ride-hailing services eg. Grab

What gets sorted:

- Patients by urgency or appointment time
- Bank transactions by date
- Flights by departure time
- Customers by queue number
- Ride requests by distance or priority

Grab

Service & Operational Systems
Time/queue/process-based applications

Retail & Commerce Systems
Shopping-related applications

foodpanda

These systems manage products, prices, and inventory.

Examples:

- Supermarkets
- Online shopping platforms eg. Shopee, Taobao
- Warehouses
- Food delivery apps eg. GrabFood, FoodPanda

What gets sorted:

- Products by price (low → high)
- Products by rating (high → low)
- Items by popularity or sales
- Stock levels (low stock first)
- Promotions / newest arrivals

Information & Knowledge Systems
Content/records-based applications

These systems organise information and media.

Examples:

- Libraries
- Schools student management systems
- News websites
- Streaming platforms eg. Netflix, Spotify

What gets sorted:

- Books by title / author / year published
- Students by name / grade
- Articles by date (latest first)
- Movies by rating / release year
- Songs by artist / title

Spotify

Constraints:

- New sorting algorithm
Cycle Sort is an in-place, unstable comparison sorting algorithm
 - Key idea: each item is placed directly in its correct position using cycles
 - It minimises the number of swaps(writes)
- Yes, cycle sort can sort both numeric and non-numeric data
- Inputting & Displaying Data Complexity
 - Inputting data - $O(n)$
 - Displaying data - $O(n)$

Why is it good for sorting grocery?

- Swapping/moving items is time-consuming
- Like physically rearranging groceries on shelves
- Thus cycle sort is ideal to minimise movement